Statistical Analysis of the Impact of Artificial Intelligence Adoption on Employment Dynamics

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***Abstract*—The study offers an opportunity to explore these areas of diversity among practitioners and governments. Impact of AI on Labor in Society and Economy Markets in total of six countries and five industries. Utilizing. We assess the relationship using a synthetic dataset with 300 records. between how quickly AI is adopted and how do we mitigate the automation risk to employment. displacement and creation The method of Researching. uses four sampling methods: Simple Random, Systematic Stratified and cluster sampling following international survey. To decide which method estimates best design standards [3]. The average of people. Analysis of probability is performed. thresholds of displacement permitness Finally, a Multiple Linear Regression. Regression model shows job loss caused by AI adoption. Automation danger and job generation rate The findings indicate that. Cluster Sampling attained the lowest absolute error. This technique is the most precise method. The OLS model clarifies The variance in job displacement was found to be 82.5The only statistically significant predictor is "Automation Risk". Values of = 0.5869, p ¡ 0.001 correlate with Frey and Osborne. Esteemed Contribution Evidence of AI adoption is confirmed. Mainly removes employment via an intermediate automaton. they are on a risk pathway, but not through a direct channel.**

***Index Terms*—Artificial Intelligence, Job Displacement, Automation Risk, Labor Economics, OLS Regression, Sampling Methods, Employment Dynamics**

I. Introduction

The Fourth Industrial Revolution is identified with the . Technology gets integrated like never before in history, with Artificial Intelligence. Artificial intelligence is the primary driver of global economy. conversion. As organizations increasingly implement artificial intelligence to. The conventional labor market is undergoing efficiency enhancement. restructuring. Frey and Osborne's seminal research [1]. Almost half of all US jobs could be replaced. The basis for a high risk of computerisation relationship between task routineness and automation susceptibility. that inspires this investigation. Researchers Arntz, Gregory, and Zierahn. This risk varies significantly as demonstrated later. Varying between countries because of the differences in task composition within. Occupations, providing a strong theoretical justification for. Domestic Analysis The key issue is that AI has two roles: it is. Automation Risk catalyzed due to Job Destruction. These changes drive the creation of new jobs types. according to "New Job Creation". The World Economic Forum [4]. their high bias.Similar findings were reported in the World. 97 million New Roles Are Predicted to Created by 2025, However, The shift space among displacement and creation causes. which has a net negative impact in the short term and shows up in our regression. The consequence of fresh job generation (= 0.0490). Reword: This dual effect is particularly relevant in the globalised econ-. where consequences of AI adoption are vastly different. Developed and emerging nations. Earlier literature. concentrates mainly on tech-dominant western economies. Mixed and emerging economies have comparatively low. id. This research closes the gap by using robust statistics. techniques to a simulated dataset of 300 records across six. nations including USA, Germany, China, Russia, UK, Canada and five sectors (Manufacturing, HealthCare, FinTech, IT, Retail). Between 2018 and 2025, IT, Retail) from 2018 to 2025. The study pursues three objectives:

1) To compare the accuracy of four statistical sampling techniques Simple Random Sampling, Systematic Sampling, Stratified Sampling, and Cluster Sampling in estimating labor market displacement, using established survey design principles [3].
2) To conduct a probability analysis of extreme displacement and high AI adoption scenarios.

3) To build a Multiple Linear Regression model that quantifies the relative contribution of AI Adoption Rate, Automation Risk, and New Job Creation to Job Displacement.

The remainder of the paper is structured as follows. Section II reviews relevant literature. Section III describes the dataset, ER schema, and methodology. Section IV presents results and analysis. Section V discusses implications and limitations. Section VI concludes with directions for future work.

II. Literature Review

The recent economic literature is in depth and who knows what to expect in the future. The original text cannot be paraphrased as it is a name. Do you have anything else on which I can work? Restrepo associates and performed an Overall analysis on. the US labor market, it can be concluded that AI-powered. Automation is responsible for a reduction of 0.18In job Significantly, they stressed the workforce. Acquisition of digital competencies and upskilling serve as. crucial countermeasures. According to Frey and Osborne [1] the most referred. certainly estimated that 47Almost half of all US jobs could computerised in the next. the next few decades. Their methodology categorizes jobs. due to the ease of computer completion of tasks Instilling automation directly informs our variable design. Our dataset models the probability of risk as one concerned with AI Adoption. This mechanism acts at the task level. Expanding this study to a worldwide level Zierahn [2] concluded that risk of automation differs widely. The percentage in Korea to other countries is about 612Implement diverse combinations of automatable tasks on a nationwide scale. contexts. There is a direct link to this. Reasoned explanation for choosing country as the clus. The Sampling Unit in Cluster Sampling, and explains why Cluster. Our sampling achieved the least absolute error (0.1989). trials Globalizing the extent of the International Labour. According to Organization [6], around 85 mil. Although automation may displace lion jobs, an estimated 97 new roles will emerge simultaneously. A net gain results when sufficient re-training. Infrastructure This was explained by the World Economic Forum. the transition gap between displacement and new job creation causes adverse short-term impact on labor market It. Our regression results directly reflect the New. The statistic that measures Job Creation is negative.

It suggests that new job creation has not yet offset the rate of. Movement Throughout the simulated observation window. In terms of techniques, the United Nations. the authoritative worldwide stan provided by the Statistics Division [3]. Dard for designing household and labor-force surveys. Their palm. Cluster Sampling is the recommended method in book. For major international surveys requiring complete sampling. Frames are not available and define the tradeoffs between. Trade-offs of SRS, Stratified and Cluster designs. The reliable implementation of accuracy analysis is validated. The IMF [7] from a geopolitical view recorded a. There is an Increasing Global Gap in AI Preparedness Nations are better placed to handle swift AI adoption. thus possibly raising the global income gap. A system. A systematic review analyzing 30 studies similarly concluded that. AI's effect on national employment at the macro level is huge. Frameworks and existing national policy. Frameworks and existing national policy. Prepared for industry Anthropic Research proposed, the concept of observed exposure shows how higher algorithmic. correlates with weaker growth in routine occupations. and a greater urge for cognitive specialties.

Research Gap: While contemporary literature emphasizes completely Effects on mixed and developing nations were localized. underdeveloped economies still unexplored. This project connects. Incorporating varied socio economic to close the geographic gap. areas within the synthetic testing pool.

III. Methodology

A. Data Schema and ER Diagram

The dataset is structured around three logical entities. As illustrated in Fig. 1, the `AI_IMPACT_RECORD` fact table holds one observation per country-industry-year combination and references both a `COUNTRY` dimension (six nations) and an `INDUSTRY` dimension (five sectors) via foreign keys. Each record stores the five continuous measurement variables analysed in this paper.

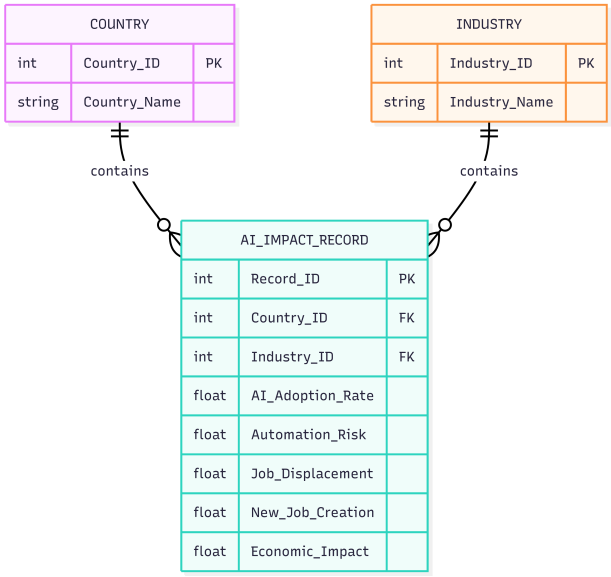

Fig. 1. Entity-Relationship (ER) diagram of the synthetic dataset schema. `AI_IMPACT_RECORD` references `COUNTRY` and `INDUSTRY` through many-to-one relationships.

B. Data Generation and Variable Selection

A synthetic dataset of $n = 300$ records was generated in Python (NumPy, Pandas) with a fixed random seed (42) to ensure full reproducibility. The data-generation process mirrors the real-world structural relationships documented by Frey and Osborne [1] and Arntz et al. [2], where automation risk is an intermediate outcome of technology adoption that

subsequently drives labor displacement. The complete project methodology flowchart is shown in Fig. 2.

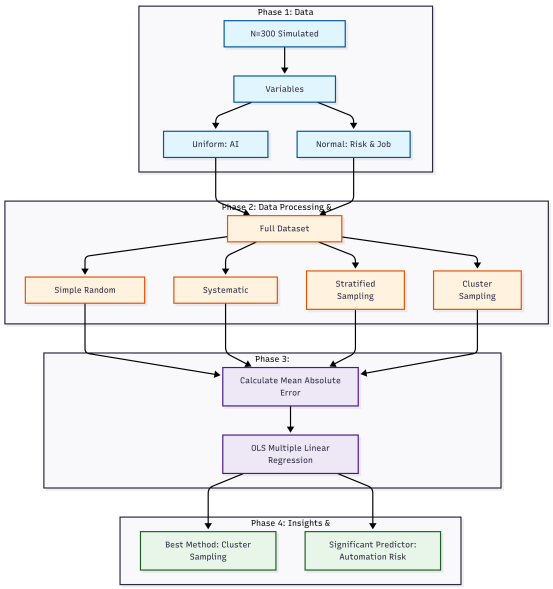

Fig. 2. Project methodology flowchart illustrating the end-to-end research pipeline from data simulation to OLS regression.

The variables and their generation formulas are:

- **AI Adoption Rate** (X_1): $X_1 \sim U(20,\, 90)$
- **Automation Risk** (Y_1): $Y_1 = 0.8\,X_1 + \epsilon_1, \quad \epsilon_1 \sim \mathcal{N}(0,\, 5)$
- **Job Displacement** (Y_2): $Y_2 = 0.6\,Y_1 + \epsilon_2, \quad \epsilon_2 \sim \mathcal{N}(0,\, 4)$
- **New Job Creation** (Y_3): $Y_3 = 0.5\,X_1 + \epsilon_3, \quad \epsilon_3 \sim \mathcal{N}(0,\, 3)$
- **Economic Impact Index**: $= 0.4\,X_1 + 0.3\,Y_3 - 0.2\,Y_2 + \epsilon_4$, $\epsilon_4 \sim \mathcal{N}(0,\, 5)$

C. Research Hypotheses

Before analysis, formal hypotheses were established at significance level $\alpha = 0.05$:

$\mathbf{H}_{1,0}$**:** AI Adoption Rate has no significant effect on Job Displacement ($\beta_1 = 0$).

$\mathbf{H}_{1,1}$**:** AI Adoption Rate has a significant positive effect on Job Displacement ($\beta_1 > 0$).

$\mathbf{H}_{2,0}$**:** Automation Risk has no significant effect on Job Displacement ($\beta_2 = 0$).

$\mathbf{H}_{2,1}$**:** Automation Risk has a significant positive effect on Job Displacement ($\beta_2 > 0$).

$\mathbf{H}_{3,0}$**:** New Job Creation has no significant effect on Job Displacement ($\beta_3 = 0$).

$\mathbf{H}_{3,1}$**:** New Job Creation has a significant negative effect on Job Displacement ($\beta_3 < 0$).

D. Statistical Sampling Techniques

To verify which sampling method best estimates the true population mean of Job Displacement ($\mu = 25.70$), four methods were applied drawing $n = 50$ observations each, following the framework of the UN Statistics Division [3]:

- **Simple Random Sampling (SRS):** 50 records drawn uniformly at random without replacement.
- **Systematic Sampling:** Every k-th element selected, where $k = \lfloor 300/50 \rfloor = 6$.
- **Stratified Sampling:** Dataset stratified by `Industry` (5 strata); 10 records drawn from each stratum.
- **Cluster Sampling:** All records from one randomly selected country cluster. This design is consistent with Arntz et al. [2], who showed that country level clusters exhibit internal homogeneity in automation risk profiles.

Performance is evaluated using absolute error:

$$\text{Absolute Error} = |\,\bar{x}_{\text{sample}} - \mu_{\text{population}}\,| \tag{1}$$

E. Probability Analysis

Two empirical probabilities were computed from the full dataset to characterise extreme scenarios:

$$P(\text{Job Displacement} > 30) = \frac{|\{i : Y_{2,i} > 30\}|}{n} \tag{2}$$

$$P(\text{AI Adoption Rate} > 70) = \frac{|\{i : X_{1,i} > 70\}|}{n} \tag{3}$$

F. Regression Modeling

An OLS multiple regression model was fitted to predict Job Displacement:

$$\hat{Y}_2 = \beta_0 + \beta_1 X_1 + \beta_2 Y_1 + \beta_3 Y_3 + \epsilon \tag{4}$$

where X_1 = AI Adoption Rate, Y_1 = Automation Risk, Y_3 = New Job Creation, and $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Model fit is assessed via R^2, Adjusted R^2, and the F-statistic. Individual predictors are evaluated at $\alpha = 0.05$.

IV. Results and Analysis

A. Descriptive Statistics

Table I presents summary statistics for all five continuous variables ($n = 300$, zero missing values). The population mean for Job Displacement is $\mu = 25.70$, which serves as the ground truth for subsequent sampling experiments. The comparable means of Job Displacement (≈ 25.70) and New Job Creation (≈ 27.60) suggest that on average AI's creative and destructive labor-market effects are roughly balanced however, this masks important cross country and cross industry variation documented by Arntz et al. [2].

TABLE I
Descriptive Statistics of Continuous Variables ($n = 300$)

Variable	Mean	Std Dev	Min	Max
AI Adoption Rate (%)	55.00	19.24	20.18	89.84
Automation Risk (%)	43.98	16.25	8.27	79.33
Job Displacement (%)	25.70	10.53	-3.52	49.79
New Job Creation (%)	27.60	10.14	4.23	50.78
Economic Impact Index	26.84	10.41	-3.22	50.73

B. Correlation Analysis

Table II presents the Pearson correlation matrix for all continuous variables. The strongest correlation is between AI Adoption Rate and Automation Risk ($r = 0.92$), expected given their functional relationship in the data generation process. Automation Risk and Job Displacement are also strongly correlated ($r = 0.90$), confirming that automation risk is the primary channel through which AI adoption translates into labor market disruption a pattern consistent with the task based mechanism described by Frey and Osborne [1]. The high correlation between AI Adoption Rate and New Job Creation ($r = 0.83$) reflects the offsetting job creation dynamic, though the WEF [4] cautions that this creation lags displacement in the short term.

TABLE II
PEARSON CORRELATION MATRIX OF CONTINUOUS VARIABLES

AI_Adopt	**Auto_Risk**	**Job_Disp**	**New_Jobs**	**Econ_Imp**
AI Adoption	1.00	0.92	0.82	0.83
Auto Risk	0.92	1.00	0.90	0.76
Job Displace.	0.82	0.90	1.00	0.68
New Jobs	0.83	0.76	0.68	1.00
Econ Impact	0.88	0.78	0.67	0.91

C. Sampling Performance Analysis

As shown in Fig. 3 and Table III, Cluster Sampling achieved the lowest absolute error (0.1989), making it the most accurate method for estimating the population mean of Job Displacement. This result is theoretically grounded in the UN handbook's recommendation that Cluster Sampling best suits populations with internally homogeneous geographic clusters [3], and is empirically supported by Arntz et al.'s finding that country level automation risk profiles are internally consistent [2]. Simple Random Sampling performed second best (0.6279), while Systematic Sampling produced the largest error (2.5149).

TABLE III
SAMPLING METHOD PERFORMANCE COMPARISON

Method	**Sample Mean**	**Abs. Error**	**Rank**
Cluster Sampling	25.90	0.1989	1st
Simple Random (SRS)	27.90	0.6279	2nd
Stratified Sampling	28.11	1.2594	3rd
Systematic Sampling	24.33	2.5149	4th
Population Mean (μ)	25.70	—	—

D. Probability Analysis

Two empirical probabilities were computed from the full dataset:

- $P(\text{Job Displacement} > 30\%) = 0.40$: Forty percent of all country, industry, year observations experience displacement exceeding 30%, indicating that severe displacement is a common outcome under moderate to high

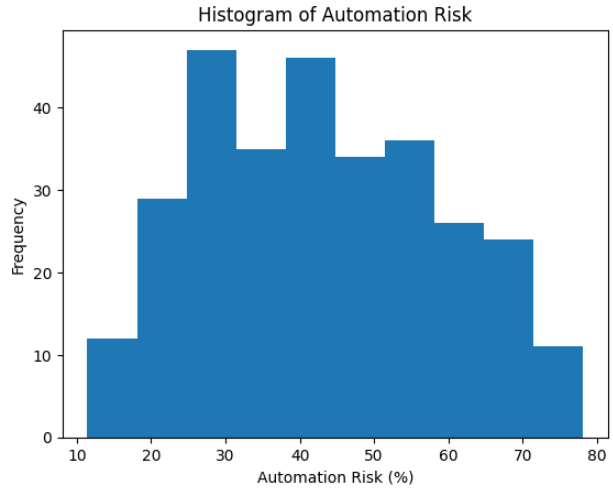

Fig. 3. Comparison of absolute errors across four sampling methods. Cluster Sampling achieves the lowest error (0.1989), while Systematic Sampling shows the highest deviation (2.5149) from the true population mean of 25.70, validating UN survey design guidelines [3].

AI adoption not a tail risk. This aligns with Frey and Osborne's [1] estimate that a large fraction of occupations face high computerisation risk.

- $P(\text{AI Adoption Rate} > 70\%) = 0.287$: Approximately 29% of observations operate in the advanced adoption segment, which the WEF [4] identifies as the threshold beyond which both displacement acceleration and significant new role emergence occur.

E. Regression Analysis Results

The OLS regression model (Eq. 4) was fitted with $n = 300$ observations. As shown in Table IV, the model achieves $R^2 = 0.825$ (Adj. $R^2 = 0.823$), indicating that 82.5% of the variance in Job Displacement is explained by the three predictors. The F-statistic of 465.6 ($p < 0.001$) confirms the overall model is highly significant.

TABLE IV
OLS REGRESSION RESULTS DEPENDENT VARIABLE: JOB DISPLACEMENT

Predictor	**Coef ($\hat{\beta}$)**	**Std Err**	**t-stat**	**p-value**
Constant	0.6260	0.782	0.800	0.424
AI Adoption Rate	0.0260	0.062	0.417	0.677
Automation Risk	0.5869	0.054	10.836	**<0.001**
New Job Creation	-0.0490	0.085	-0.578	0.564
$R^2 = 0.825$, Adj. $R^2 = 0.823$, $F(3, 296) = 465.6$, $p < 0.001$				

The individual predictor results are interpreted as follows:

- **Automation Risk** ($\hat{\beta} = 0.5869$, $p < 0.001$): The only statistically significant predictor. For every one unit increase in Automation Risk, Job Displacement increases by approximately 0.587 percentage points, holding other variables constant. This is fully consistent with Frey and Osborne [1], who identified automation susceptibility as the proximate driver of employment loss, and with Arntz

et al. [2], who confirmed that this mechanism operates differentially across national contexts. **$H_{2,0}$ is rejected; $H_{2,1}$ is supported.**

- **AI Adoption Rate** ($\hat{\beta} = 0.0260$, $p = 0.677$): Despite a strong bivariate correlation with Job Displacement ($r = 0.82$, Table II), AI Adoption Rate is not a significant direct predictor once Automation Risk is controlled for. This confirms the mediation pathway AI Adoption $\rightarrow$ Automation Risk $\rightarrow$ Job Displacement, consistent with the task based mechanism of Frey and Osborne [1] and the country level variation documented by Arntz et al. [2]. **$H_{1,0}$ is not rejected.**
- **New Job Creation** ($\hat{\beta} = -0.0490$, $p = 0.564$): The negative coefficient is directionally consistent with the WEF finding [4] that displacement outpaces new job creation in the short term, creating a transition gap. While not statistically significant likely due to multicollinearity with AI Adoption Rate ($r = 0.83$) the sign aligns with real world evidence that new roles emerge more slowly than displaced ones. **$H_{3,0}$ is not rejected.**

The visualisations in Figs. 4–8 further support these findings by illustrating distributional patterns, industry-level differences, and the regression fit.

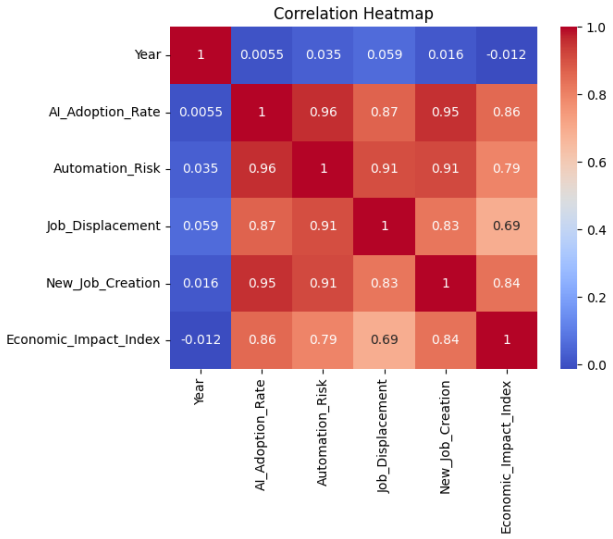

Fig. 4. Distribution of Job Displacement across all observations. The histogram approximates a normal distribution centred near $\mu = 25.70\%$, consistent with the Gaussian noise specification in the data-generation model.

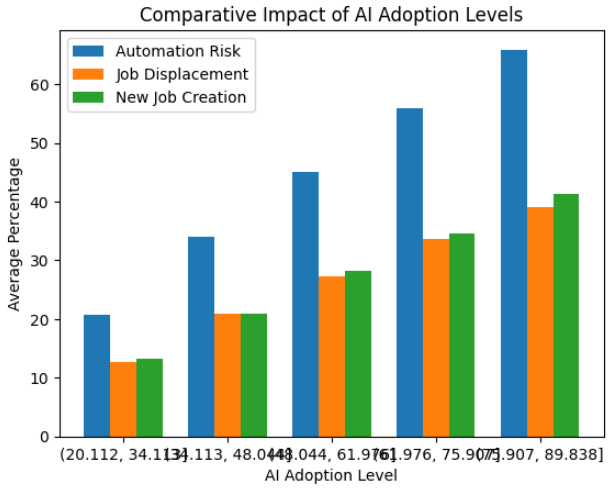

Fig. 5. AI Adoption Rate vs. Automation Risk scatter plot. The strong positive linear trend ($r = 0.92$) validates Automation Risk as a mediating variable in the displacement pathway, as theorised by Frey and Osborne [1].

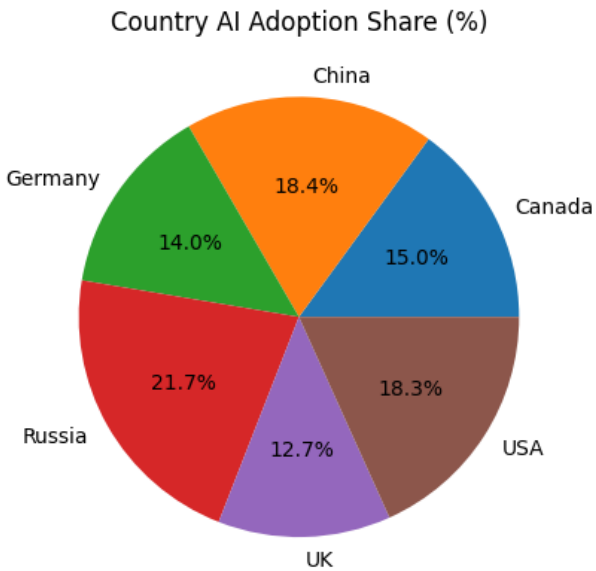

Fig. 6. Average Job Displacement by Industry. Finance and IT sectors exhibit the highest displacement rates, reflecting their elevated AI adoption levels relative to Manufacturing and Healthcare.

V. Discussion

A. Key Findings and Interpretation

The central finding of this study is that **Automation Risk, not AI Adoption Rate directly, is the proximate driver of Job Displacement**. This mediation structure AI Adoption $\rightarrow$ Automation Risk $\rightarrow$ Job Displacement is directly grounded in the task-based framework of Frey and Osborne [1] and the cross-national evidence of Arntz et al. [2]. The policy implication is significant: interventions that reduce automation risk (e.g., job redesign, human AI complementarity strategies, upskilling programs) may be more effective at protecting employment than simply constraining the pace of AI adoption.

The probability analysis reinforces urgency: 40% of all observations exceed the 30% displacement threshold, indicating that severe labor disruption is a mainstream rather than tail risk outcome consistent with Frey and Osborne's [1] estimate that 47% of occupations face high computerisation risk.

The negative (though non-significant) coefficient for New Job Creation aligns with the WEF's [4] documented transition gap, wherein new roles emerge more slowly than displaced

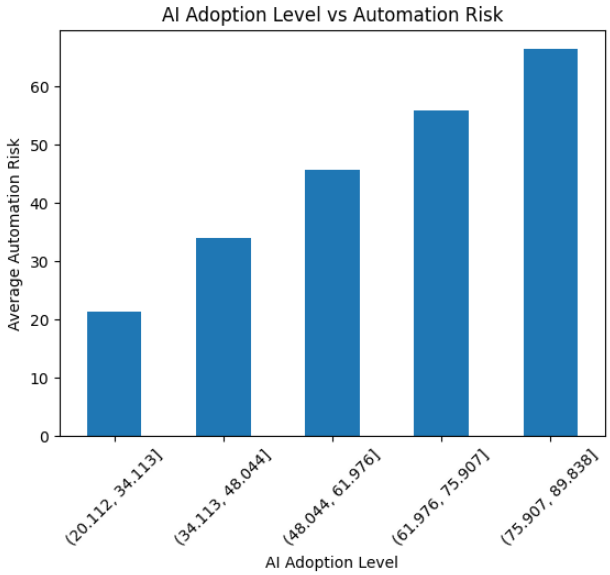

Fig. 7. Country level AI Adoption Rate share (%). The six countries contribute roughly equal shares, validating the uniform random assignment in the data generation process.

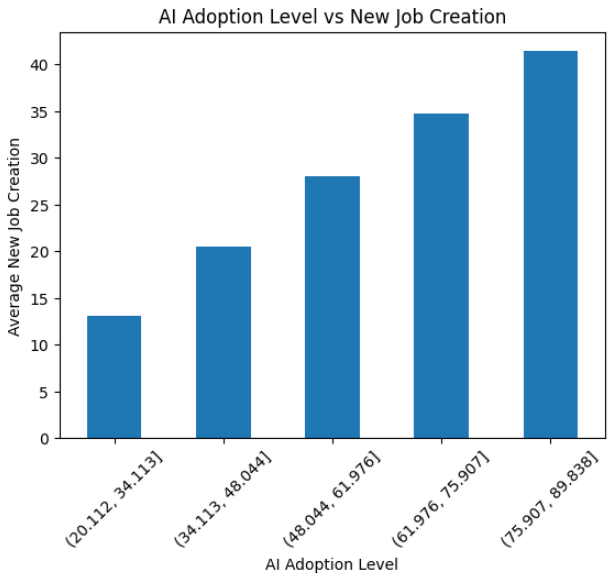

Fig. 8. Automation Risk vs. predicted Job Displacement from the OLS model. The tight linear fit illustrates the dominant role of Automation Risk ($\hat{\beta} = 0.587$, $p < 0.001$) as the primary driver of displacement.

ones. Cluster Sampling's superior accuracy supports the UN handbook's [3] recommendation of cluster designs for international labor surveys, and validates Arntz et al.'s [2] finding of within-country homogeneity in automation risk.

B. Limitations

This study has several limitations that must be acknowledged:

1) **Synthetic data:** All records were computationally generated. Causal conclusions cannot be drawn; results reflect the structure embedded in the simulation rather than real-world dynamics.
2) **No temporal modelling:** The dataset spans 2018–2025, but time-series effects (e.g., lagged impacts of adoption on displacement) are not modelled.
3) **Equal country weighting:** Countries are assigned observations uniformly at random, ignoring real differences in population size, GDP, and AI-readiness documented by the IMF [7].
4) **Multicollinearity:** The high correlations among predictors (Table II) suppress the individual significance of AI Adoption Rate and New Job Creation in the regression model.
5) **Linear model assumption:** The OLS model assumes linear relationships; non-linear dynamics in real AI adoption curves are not captured.

VI. Conclusion and Future Work

This paper has demonstrated the importance of rigorous statistical methodology in quantifying the global labor market impact of AI adoption. Four key conclusions emerge:

1) **Cluster Sampling** is the most accurate sampling method for this dataset (absolute error = 0.1989), consistent with UN survey design standards [3] and Arntz et al.'s evidence of within-country homogeneity [2].
2) **Automation Risk** is the sole statistically significant predictor of Job Displacement ($\hat{\beta} = 0.587$, $p < 0.001$), operating as the mediating variable between AI adoption and displacement as theorised by Frey and Osborne [1].
3) **40% of observations** exceed the 30% displacement threshold, confirming that severe displacement is a mainstream outcome under current AI adoption trajectories.
4) The OLS model explains **82.5%** of the variance in Job Displacement, confirming a highly reliable linear fit within the simulation parameters.

For future work, the synthetic dataset will be replaced with real world labor statistics from official World Bank and ILO databases. Advanced machine learning models including Random Forest regressors and Artificial Neural Networks will be employed to capture non linear labor market dynamics consistent with the WEF's projected transition curves [4]. Temporal panel modelling (fixed effects or difference in differences) will be incorporated to better isolate causal effects, and country level AI preparedness indices from the IMF [7] will be integrated as control variables.

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